

Program Question_3

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Implicit none
Real::x0,x_new,f_val,f_deriv,relative_error
Integer::iter,max_iter=100
Real,Parameter::tol=1.0E-5
!Initial Guess
Call Newton_Method(-5.0)
Call Newton_Method(-2.0)
call Newton_Method(1.0)
call Newton_Method(4.0)

contains
Function f(x) Result(res)
  Real,Intent(In)::x
  Real::res
  res=5*x**2+cos(3*x)-2*exp(x)-exp(-x)
end function f
Function f_prime(x) Result(res)
  Real,Intent(In)::x
  Real::res
  res=10.0*x-3.0*sin(3*x)-2.0*exp(x)+exp(-x)
end function f_prime
subroutine Newton_Method(x0)
  Real,Intent(In)::x0
  Real::x_new,x_prev,f_val,f_deriv,relative_error
  Integer::iter
  x_prev=x0
  iter=0
  relative_error=1.0
  Write(*,*)"Iital Guess x0=",x0
  Write(*,*)"Iter   |P(n-1)   |f'(P(n-1))   |f(P(n))   |f(P(n))   Relative_Error "

  Do while(relative_error>tol .And. iter<100)
    f_val=f(x_prev)
    f_deriv=f_prime(x_prev)
    If(abs(f_deriv)<1.0E-10)then
      Write(*,*)"Derivative Too small , so newton method Fail"
      Exit
    End If
    x_new= x_prev -(f_val/f_deriv)

    relative_error=abs((x_new-x_prev)/x_new)
    Write(*,'(I4,3F12.6,3F12.6)')iter,x_prev,f_val,f_deriv,x_new,f(x_new),relative_error
    x_prev=x_new
    iter=iter+1

  End Do
  Write(*,*)"The Aproximation is x_new=",x_new,"after",iter,"Iteration"
End Subroutine
```

End Program